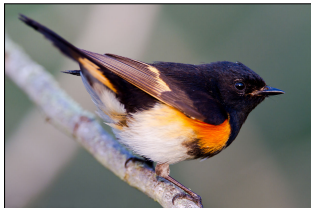


Age-structured and stage-structured population models



TODAY'S TOPICS

- 1 INTRODUCTION
- 2 MATRIX MODELS
- 3 REPRODUCTIVE VALUE
- 4 STAGE-STRUCTURED MODELS

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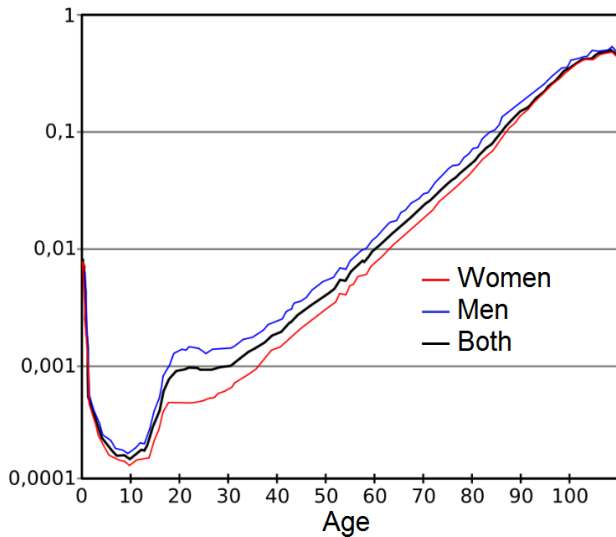
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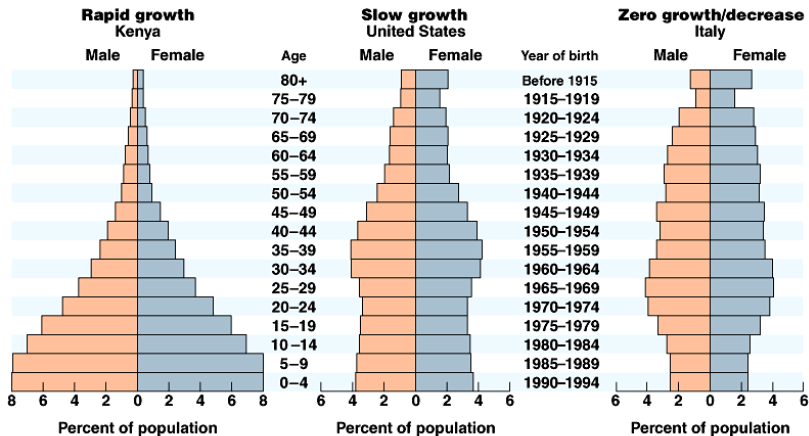
These facts have important management implications.

HUMAN MORTALITY RISK



HUMAN AGE DISTRIBUTIONS

Declining populations have relatively more old individuals than growing populations



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THREE AGE CLASS EXAMPLE

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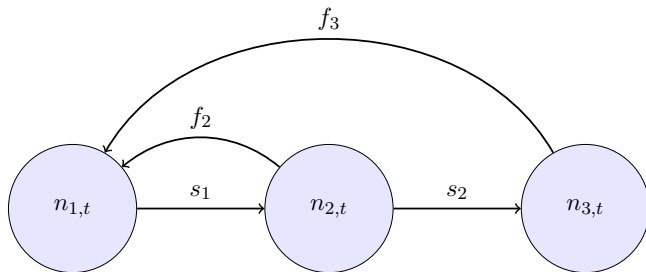
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HOW DOES THIS POPULATION GROW?

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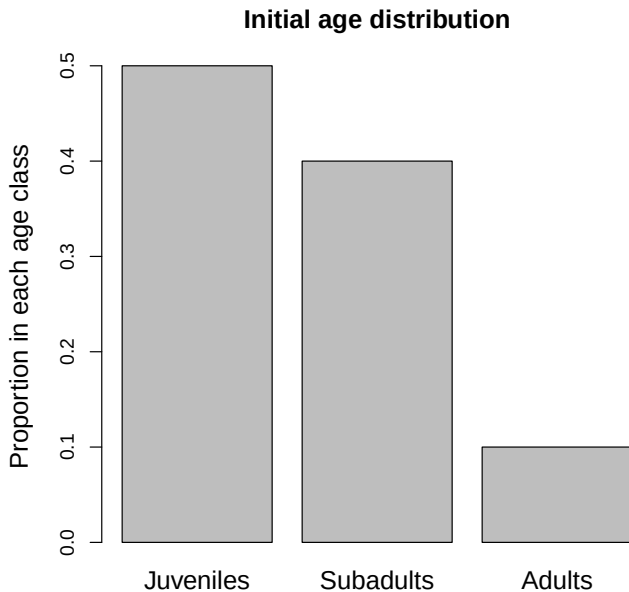
In stage-structured models, animals can stay in a class for more than one time interval.

- Suppose initial abundance in the 3 age classes is:
 - ▶ $n_{1,1} = 50$ (Age class 1, juveniles)
 - ▶ $n_{2,1} = 40$ (Age class 2, subadults)
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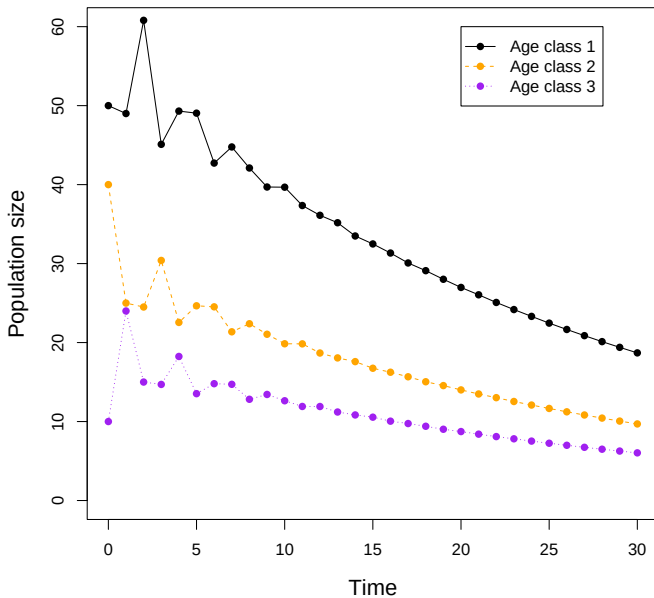
An age distribution describes the proportion of individuals in each age class



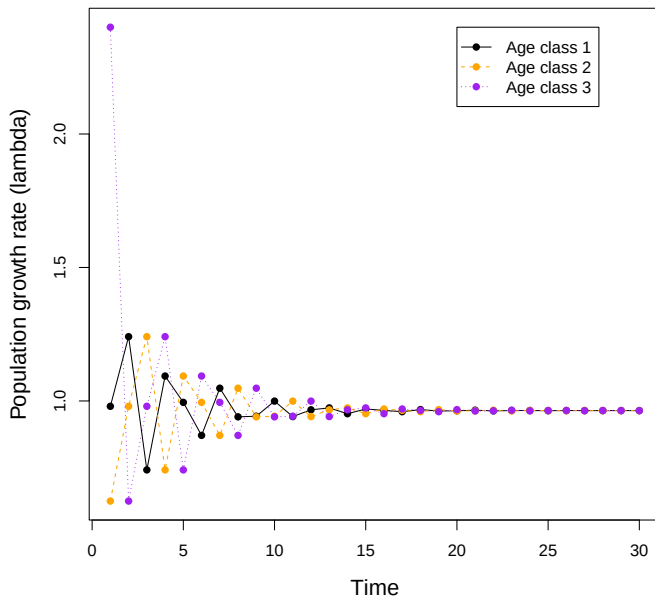
Let's choose values of s_i and f_i

Age class	Initial population size	Survival probability	Fecundity rate
	$n_{i,1}$	s_i	f_i
1	50	0.5	0.0
2	40	0.6	0.8
3	10	0.0	1.7

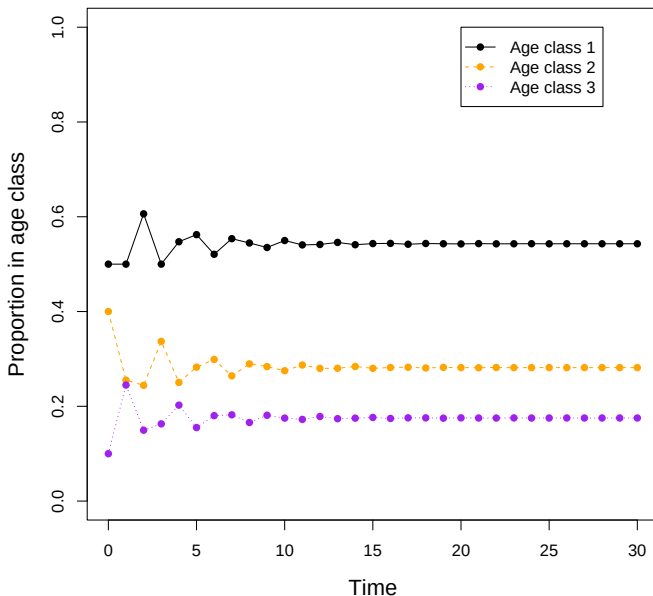
POPULATION SIZE, $n_{i,t}$



GROWTH RATES, $\lambda_{i,t} = n_{i,t}/n_{i,t-1}$



AGE DISTRIBUTION, $c_{i,t} = n_{i,t}/N_t$



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Asymptotic growth rate is λ (without subscript).

Growth rate at the stable age distribution is the same for all age classes, and it is geometric!

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But, they make it much easier to compute important quantities like λ and *reproductive value*.

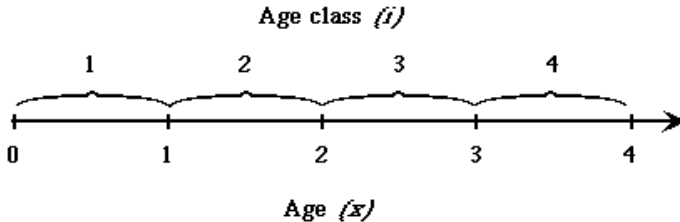
MATRIX MODELS VS LIFE TABLES

Matrix models

- Age is discrete

Life tables

- Age is continuous



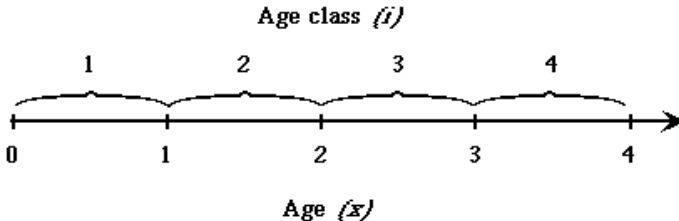
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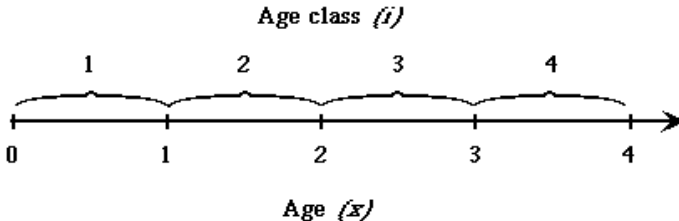
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WHAT IS A MATRIX?

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Example of a 3×4 matrix:

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} \end{bmatrix}$$

What is it?

- Square matrix
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Example:

$$\mathbf{A} = \begin{bmatrix} f_1 & f_2 & f_3 & f_4 \\ s_1 & 0 & 0 & 0 \\ 0 & s_2 & 0 & 0 \\ 0 & 0 & s_3 & 0 \end{bmatrix}$$

HOW DO WE USE A LESLIE MATRIX?

These two expressions are equivalent:

Age class	Equation
1	$n_{1,t+1} = n_{1,t} \times f_1 + n_{2,t} \times f_2 + n_{3,t} \times f_3$
2	$n_{2,t+1} = n_{1,t} \times s_1$
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AND

$$\mathbf{n}_{t+1} = \mathbf{A} \times \mathbf{n}_t$$

$$= \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix} \times \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix}$$

$$\begin{bmatrix} aw + bx + cy + dz \\ ew + fx + gy + hz \\ iw + jx + ky + lz \\ mw + nx + oy + pz \end{bmatrix} = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix} \times \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix}$$

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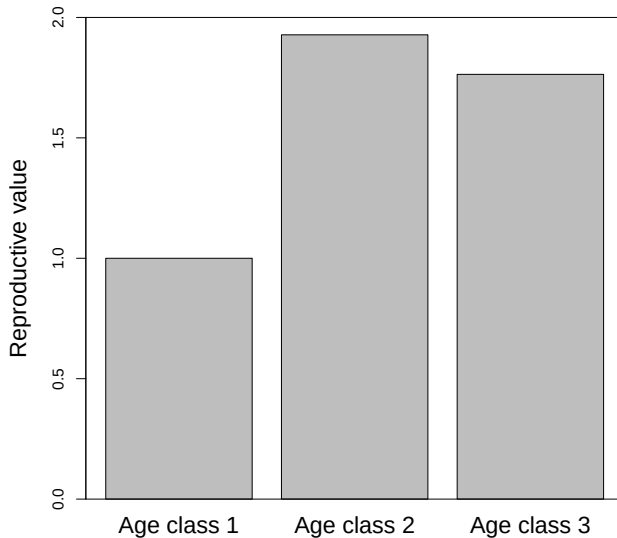
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Post-reproductive individuals have reproductive values of zero.

Depending on survival and fecundity rates, a first-year individual could have a higher or lower reproductive value than a second-year individual.

REPRODUCTIVE VALUE FOR PREVIOUS EXAMPLE



Net reproductive rate

The expected number of individuals produced by an individual over its lifetime:

$$R_0 = \sum_{i=1}^I \left(\prod_{j=1}^{i-1} s_j \right) f_i$$

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Generation time

The time required for a population to increase by a factor of R_0 .

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¹This is for graduate students only

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OTHER PROPERTIES OF THE LESLIE MATRIX¹

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The right eigenvector is the **stable age distribution**.

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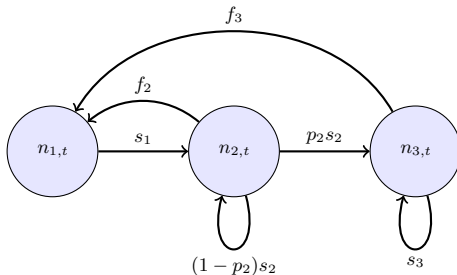
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This example allows for individual to remain in stages 2 and 3 for more than one time period.

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Matrix models are a convenient method used to work with age-structured populations.